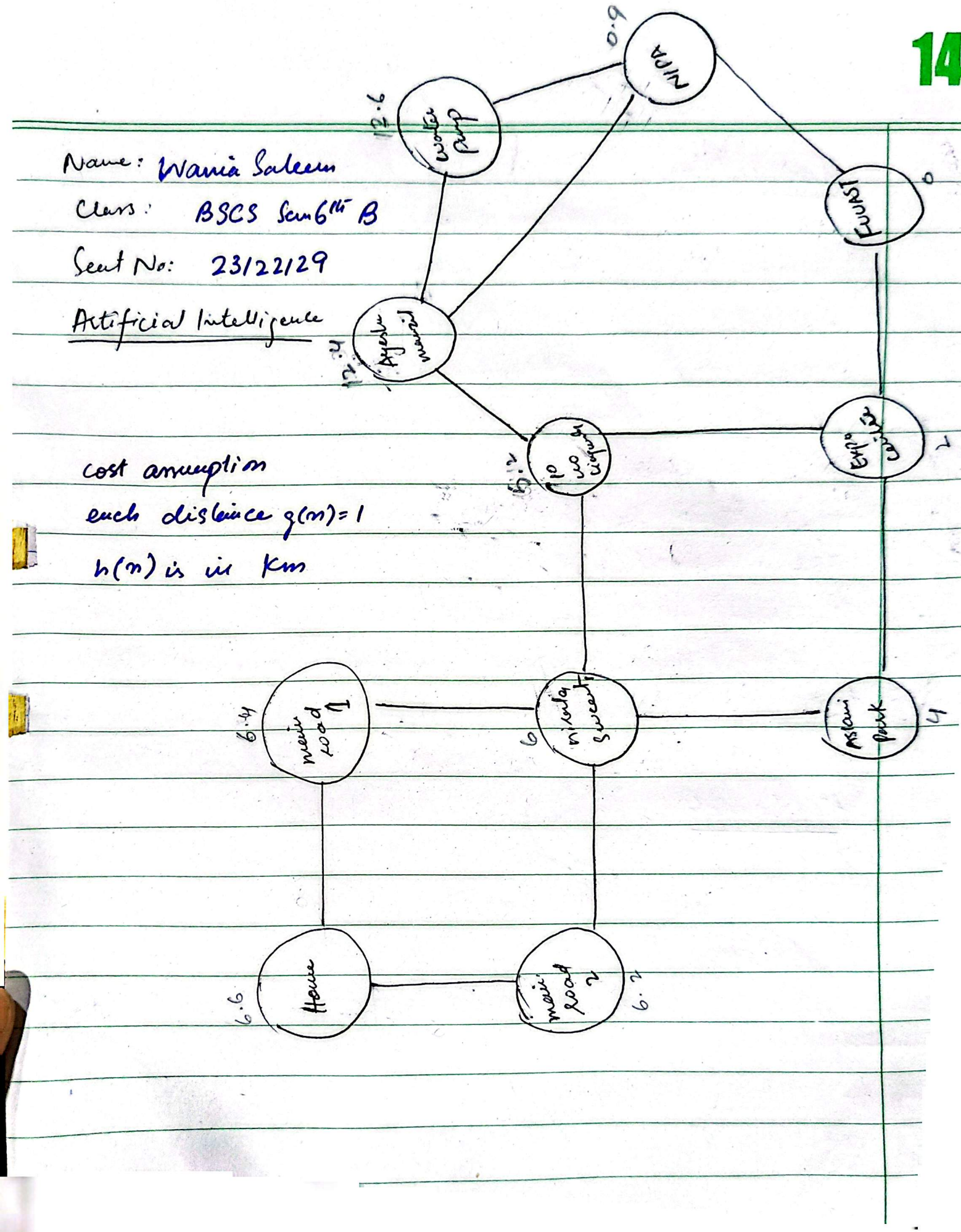
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Class: BSCS 6th B

Seat No: 23122129

Artificial Intelligence



cost assumption

each distance $g(n) = 1$

$h(n)$ is in Km

Translation

- Home → mainroad1, mainroad2
- mainroad 1 → Home, Niala
- mainroad 2 → Home, Niala
- Niala → mainroad 1, mainroad 2, Askari Park, 10 no.
- Askari Park → Niala, expo centre
- 10 no → Niala, expo centre, Ayesha manzil
- expo centre → Askari Park, 10 no., FUUAST
- Ayesha manzil → 10 no. NIPA, malirpump
- malirpump → Ayesha manzil, NIPA
- NIPA → Ayesha manzil, malirpump, FUUAST
- FUUAST → expo centre, NIPA

A* Search ∴ $f(n) = g(n) + h(n)$

Step: 0

- PQ : [Home]
  V : [ ]
- $f(n) =$ ~~0~~ $0 + 6.6 = 6.6$
- Parent table :

  Home ← None

Step : 1

- expand Home, Home → mainroad 1 mainroad 2
- PQ : [mainroad 1, mainroad 2]
  V : [Home]
- f(n) = mainroad 1 = 1 + 6.4 = 7.4
  mainroad 2 = 1 + 6.2 = 7.2
  choose mainroad 2
- Parent table:
  Home ← None
  mainroad 2 ← Home

Step : 02

expand mainroad 2 → Home, Nivala sweets.

- PQ : [mainroad 1, ~~mainroad~~ Nivala sweets]
  V : [Home, mainroad 2]
- f(n) = 2 + 6 = 8
  choose Nivala sweets
- Parent table:
  Home ← None
  mainroad 2 ← Home
  Nivala Sweets ← mainroad 2

CamScanner

Step 03

- expand Nivala sweets → Askari Park, 10. No., mainroad1, mainroad2.
- PQ : [mainroad1, Askari Park, 10 no.]
  V: [Home, mainroad2, Nivala Sweets]
- f(n) → Askari Park = 3 + 4 = 7
  10 no = 3 + 5.2 = 8.2
  choose Askari Park route
- Parent Table:

  Home ← None
  mainroad2 ← Home
  Nivala Sweets ← mainroad2
  Askari Park ← Nivala Sweets.

Step 04

expand askari Park → Nivala Sweets, expo centre

- PQ : [mainroad1, 10 no., expo centre]
  V: [Home, mainroad2, Nivala sweets, Askari Park
- f(n) = expo centre : 4 + 2 = 6
- parent table:

  Home ← None
  mainroad2 ← Home
  Nivala Sweets ← mainroad2
  Askari Park ← Nivala Swe
  expo centre ← Nivala Sw

Step 05

expand expo centre → 10 no, Askari Park, FUUAST

- PQ : [mainroad1, 10 no, FUUAST]
  V: [Home, main road2, Nivala Sweet, Askari Park, expo centre]
- f(n) = 5 + 0 = 5 → FUUAST
- Parent Table

Home ← None

mainroad2 ← Home

Nivala Sweets ← mainroad2

Askari Park ← ~~no~~ Nivala Sweets

expo centre ← Askari Park

FUUAST ← expo centre.

